Status	Finished
Started	Wednesday, 21 October 2026, 2:01 PM
Completed	Wednesday, 21 October 2026, 2:14 PM
Duration	13 mins 18 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging

b. Searching

c. Complexity

d. Sorting

The correct answer is:
Complexity

Question **2**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are merged without sorting

b. Each part is sorted independently

c. The parts are ignored

d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Bubble Sort
- b. Merge Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 40,80,90,150,200,300,400,470,70,120
- d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Insertion Sort
- c. Bubble Sort
- d. Quick Sort

The correct answer is: Bubble Sort

Question **6**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort cannot handle duplicate elements
b. Bubble Sort is less efficient for large lists
c. Bubble Sort does not guarantee sorted order
d. Bubble Sort is difficult to implement

The correct answer is: Bubble Sort is less efficient for large lists

Question **7**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Using the divide-and-conquer approach
b. Being the most efficient sorting algorithm
c. Bubbling up the largest element to its correct position with each pass
d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting is rarely used in programming
b. Sorting helps in finding duplicates quickly
c. Sorting decreases the efficiency of selection operations
d. Sorting makes it harder to search for items

The correct answer is: Sorting helps in finding duplicates quickly

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Finding a specific element in a list
- b. Arranging data in a specific order
- c. Merging two datasets
- d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order

Question **10**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Duplicates
- b. Selection
- c. Searching
- d. All of the above

The correct answer is: All of the above

Question **11**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sort()
- b. sorted()
- c. order()
- d. arrange()

The correct answer is: sort()

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Combining sorted sublists
- c. Sorting the entire list sequentially
- d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **13**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Greedy method
- b. Dynamic programming
- c. Backtracking
- d. Divide and conquer

The correct answer is:
Divide and conquer

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Bubble Sort
- b. Heap Sort
- c. Quick Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search
b. Merge Sort
c. Binary Search
d. Bubble Sort

The correct answer is: Merge Sort